

8 (a) (i) most alphas have small deflection

so nucleus is small target

(ii) deflection too large to be gravitational

so must be electrostatic i.e. charged

- (b) (i) 1. from the graph, $Q = 1.2 \text{ MeV}$
2. $v = \sqrt{E/(1/2 m_e)}$
 $= 6.5 \times 10^8 \text{ ms}^{-1}$

3. The calculated value of v is more than c , the speed of light.
(This suggests that the classical formula for kinetic energy $\frac{1}{2}mv^2$, is not valid in calculating the speed of the beta particle.)

(ii) $m(\text{Po}) = m(\text{Bi}) - m_e - Q/c^2$
 $= [(209.939 \text{ u}) - (9.11 \times 10^{-31} / 1.66 \times 10^{-27}) - (1.2 \times 10^6 \times 1.60 \times 10^{-19} / c^2)] / (1.66 \times 10^{-27} \text{ kg})$
 $= 209.937 \text{ u}$

(iii) Range of values accepted : 0.16 - 0.18 MeV

(iv) Since ^{210}Bi undergoes spontaneous beta decay, a process in which it increases its proton number by one while decreasing its neutron number by one, it suggests that ^{210}Bi must have an excess of neutrons as compared to the optimal ratio.

(v) For a stationary nucleus decaying into the beta particle and daughter nucleus, the conservation of linear momentum requires that $\underline{p_1} = -\underline{p_2}$.
The sum of kinetic energies will thus be $(p_1)^2/2m_1 + (p_2)^2/2m_2 = E$, which ought to equal the energy released in the reaction, which, if equal to the increase in the total binding energy/decrease in total mass, ought to be constant.

The range of beta particle energies and thus the supposed energy released E , seem to suggest that the energy released was not constant, in contradiction to the principle of conservation of energy.

- (vi) • Energy was not conserved in a beta decay. (proven false)
• **The existence of another undetected particle.**
• (The actual reason. The undetected particle was the neutrino, and a 3 body interaction allowed for the beta particle to carry away a varying amount of KE whilst still ensuring COE)

(b) (i) random nature of emissions

(ii) e.g. self-absorption, detector not 100% efficient,
detector not surrounding source

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$$R = R_0 e^{-\lambda t} \text{ so } 8.3 \times 10^3 = 7.6 \times 10^8 e^{-4.6 \times 10^{-3} t} \quad t = 2480 \text{ s}$$

ii Using inverse square law, answer is 26 counts per minute.